

the customer's measurement needs. They also manage and grow customer base for the company. An application engineer has in-depth knowledge of measurement techniques and products, and is responsible for demonstrating and applying solutions, mostly on-site at the customer's place. This enables sales and increases customers' confidence in the company and its capabilities."

"Marketing engineers execute all marketing-related initiatives to enhance sales activities. They are responsible for introduction of new products and technology in the marketplace and constantly assessing customer needs. Service engineers provide post-sales repair and calibration support to the end-users and help maintain their product uptime and precision," she adds.

Other opportunities are in the areas of automation and control, plant maintenance, instrumentation design, sales and marketing, service engineering and R&D (DRDO, ISRO and DAE).

"Instrumentation engineers may be required to travel extensively and frequently to remote areas for equipment installation or maintenance of already installed equipment," informs Kamat. "R&D and design requires a deep insight into analogue electronics, which is a rare skill these days in fresh

Job prospects (Courtesy: <http://education.sulekha.com/>)

- Instrumentation engineers find jobs in R&D units in both private and public sectors.
- Opportunities are available in heavy industries such as steel plants, thermal power stations, refineries, fertiliser plants and cement plants. Instrumentation engineers are also employed by defense establishments.
- Career avenues are available in management, sales and technical projects.
- Top employers for instrumentation engineers in India:
 - (a) Private sector: UltraTech Cement, Alchem International, A.R. Software, HCL Infosystems, TCS, Infosys, Accenture
 - (b) Public sector: Railway Recruitment Board, Steel Authority of India Ltd (SAIL), Bharat Heavy Electricals Ltd (BHEL), National Fertilisers Ltd (NFL), Indian Oil Corporation Ltd (IOCL)

graduates from Indian universities," he adds.

Skills & training

"Apart from a training in electronics (digital and analogue), instrumentation engineering aspirants should undergo training in handling, calibration and field diagnostics of instruments in various domains," informs Kamat. Generally, this training is provided by the employers to graduate trainees.

"Instrumentation engineers need to be strong in analytical and logical skills apart from being competent in their core area of academic training," Kamat says.

Talking about the training provided at his company, Kamat says, "Being an embedded systems service provider, my company imparts training in fundamental concepts of embedded systems, microcontrollers and digital signal processors and design of em-

bedded systems for instrumentation. We have a comprehensive assimilation programme for experienced newly hired as well as freshers."

Talking about the training at Agilent, Jyotsna shares, "Post the initial company- and business-related orientation, the programme is customised to the new hire. Freshers go through a three to six months long programme where they get trained in measurement science principles, products, technology and processes by subject matter experts. In addition, they get an opportunity to shadow a senior at work for a set period of time so that they can get first-hand experience of the job."

Pay package

Typically, the pay package depends on various parameters such as the size and origin of the company, criticality of the job profile and location. However, a fresh graduate may expect anywhere from ₹ 20,000 to ₹ 24,000 per month, informs Kamat.

"As with any other industry, pay packages are competitive, and in line with the candidate's qualifications and previous experience," shares Jyotsna. ●

The author is a tech correspondent at EPY Bengaluru

Top institutions for instrumentation engineering

- Most of the IITs and NITs
- Birla Institute of Technology and Science (BITS) Pilani
- Rashtriya Vidyalyaya College of Engineering
- Vellore Institute of Technology
- PSG College of Technology
- International Institute of Information Technology Bangalore
- Netaji Subhas Institute of Technology (NSIT)

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